

RetailBot Pro: A Framework for Resource-Efficient, Domain-Adapted Conversational AI in Retail via QLoRA Fine-Tuning

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I. ABSTRACT

- The rise of Large Language Models (LLMs) has opened new avenues for building domain-specific conversational systems that offer precise, context-aware insights within specialized sectors. This paper introduces RetailBot Pro, a comprehensive conversational AI tailored specifically for the retail and e-commerce industry. Our approach leverages Quantized Low-Rank Adaptation (QLoRA) used to fine-tune the TinyLlama-1.1B-Chat model — on a structured retail customer support dataset sourced from HuggingFace.
- To fine-tune the model, we utilized the Bitext customer support LLM chatbot training dataset and applied 4-bit NF4 quantization, successfully conducting training on consumer-grade hardware (Google Colab T4 GPU). The fine-tuned model is deployed as an interactive Gradio chatbot capable of handling retail queries including order tracking, return procedures, refund policies, and shipping enquiries. Experimental results indicate that this domain-adapted approach results in high response accuracy, minimal

hallucination, and sub-3-second latency.

II. INTRODUCTION

2.1 Background and Motivation

- The evolution of transformer-based architectures has significantly redefined the boundaries of human–computer interaction in modern commercial environments. Over the past decade, conversational systems have progressed from rigid, rule-based chatbots to sophisticated AI-driven assistants capable of maintaining natural, context-aware dialogue across a wide range of customer service scenarios. Large Language Models (LLMs) now enable businesses to provide scalable, always-available support, reducing operational costs while improving user engagement and accessibility. This shift has opened new opportunities for automating customer communication in industries where rapid response times and personalized interactions are critical.
- Despite these advancements, general-purpose language models often struggle when applied directly to the retail and e-commerce domain. A key limitation is the phenomenon of

hallucination, in which models generate responses that appear confident and fluent but are factually incorrect or unsupported. This issue becomes particularly problematic in retail scenarios involving order tracking, return eligibility, warranty coverage, refund policies, and shipping logistics. When customers rely on automated systems for such sensitive and time-critical information, inaccurate responses can lead to confusion, dissatisfaction, and loss of trust. Research has shown that hallucinations remain a persistent challenge for generative models, particularly when they operate outside the scope of their training data.

- The retail industry demands a significantly higher standard of accuracy and reliability than many other conversational domains. Customers expect immediate and precise information regarding their purchases, shipping timelines, product availability, and service entitlements. Even minor inaccuracies can result in negative customer experiences, increased support costs, and potential reputational damage for businesses. This creates a fundamental mismatch between the generalized capabilities of off-the-shelf language models and the rigorous requirements of retail stakeholders.
- Consequently, there is a strong need for specialized conversational AI systems that combine the fluency of generative

models with the precision of domain-specific knowledge. Such systems must be capable of grounding their responses in verified retail policies and operational workflows while maintaining the flexibility and conversational quality that modern users expect. Bridging this gap serves as a primary motivation for the development of domain-adapted conversational agents designed specifically for retail and e-commerce environments.

2.2 Research Problem

- Current retail AI tools generally fall into two categories: generic LLM interfaces that lack deep domain expertise, or static FAQ systems that lack conversational depth. Neither fully meets the needs of retail customers who require nuanced, contextually grounded explanations. Additionally, the massive computational overhead required to fully fine-tune state-of-the-art models often makes specialized development inaccessible for small-scale businesses.
- This study seeks to bridge this gap by addressing a core technical question: Is it possible to build a high-performing, retail-relevant AI system on commodity hardware by leveraging QLoRA-based parameter-efficient fine-tuning on a structured customer support dataset?

2.3 Research Objectives

- To address this problem, the study focuses on the following three

objectives:

- **Pipeline Development & Evaluation:** To build and assess a QLoRA-based fine-tuning workflow for the TinyLlama-1.1B-Chat model using a specialized retail customer support dataset, demonstrating that effective domain adaptation is possible on limited hardware.
- **Hallucination Reduction:** To implement structured prompt engineering and low-temperature generation strategies that minimize hallucination and improve factual accuracy in retail-specific responses.
- **System Deployment:** To engineer an interactive conversational interface via Gradio capable of handling real-world retail queries in a production-like environment.

2.4 Scope and Significance

The scope of this research covers the entire development cycle of RetailBot Pro, ranging from dataset curation and fine-tuning model to the final deployment of the chatbot. The system addresses a wide spectrum of retail topics, including order management, return and refund procedures, shipping enquiries, and product support.

Beyond its immediate application, this work is significant because it provides a reproducible, cost-effective blueprint for building domain-specific AI. This framework can be adapted by researchers in other customer-facing industries such as logistics, hospitality,

or banking, where precision is critical but computing budgets are limited.

III. INDUSTRY ANALYSIS

3.1 Overview of the Retail and E-Commerce Landscape

The global e-commerce market is experiencing unprecedented growth, with its valuation expected to surpass \$7.9 trillion by 2027. AI-driven customer support has emerged as a primary driver for operational efficiency and customer satisfaction. As retail delivery shifts toward digital-first environments, there is a growing demand for tools that go beyond static FAQs or scripted chatbots. Traditional support structures are no longer sufficient for modern online shoppers who expect immediate, accurate, and personalized responses at any hour of the day.

3.2 AI Adoption Challenges in Retail

While the potential for AI in retail customer support is high, several hurdles prevent seamless adoption:

- **The Risk of Hallucination:** General-purpose models often generate incorrect product or policy claims, which can mislead customers and erode trust in the brand's digital presence.
- **Specialized Vocabulary:** Retail discourse is rich with domain-specific terms — such as SKU identifiers, SLA windows, RMA procedures, and chargeback policies — that models must be specifically tuned to understand and apply accurately.
- **Hardware Barriers:** The high cost of

training large-scale models often excludes smaller retailers. Without access to massive cloud budgets, many organizations are locked out of custom AI development.

- **Lack of Continuity:** Retail support often requires multi-turn dialogue. Many basic systems lack session coherence, making them unable to track the context of an evolving customer complaint.

3.3 The Rise of AI-Augmented Retail Tools

Recent breakthroughs in Parameter-Efficient Fine-Tuning (PEFT) specifically LoRA and QLoRA — have democratized AI development for specialized domains. These techniques allow developers to customize billion-parameter models using modest hardware without sacrificing conversational quality. For the retail sector, this means the vision of an always-on, expert-level digital support agent is finally becoming a practical, cost-efficient reality.

3.4 Business Value and Industry Requirements

The implementation of domain-specific retail AI provides measurable advantages across several key areas:

- **24/7 Customer Support:** Continuous access to high-quality retail assistance without human intervention, reducing average handle time and support costs.
- **Operational Efficiency:** A significant reduction in the administrative burden on human support agents, who are often overwhelmed by repetitive policy or order queries.

- **Global Scalability:** The ability to provide high-quality, standardized support to customers across diverse geographies and time zones.
- **Brand Consistency:** Delivery of consistent, policy-aligned responses that reduce the risk of misinformation and protect brand reputation.

For retail organizations and e-commerce platforms, these systems are more than just “chatbots”; they represent a fundamental upgrade to the infrastructure of customer service delivery.

IV. LITERATURE REVIEW

4.1 Overview

Since the landmark introduction of transformer architectures by Vaswani et al. (2017), the body of research regarding specialized conversational AI has grown exponentially. This review synthesizes current scholarships across three critical areas: Parameter-Efficient Fine-Tuning (PEFT) techniques, hallucination mitigation strategies, and the practical application of AI within customer support environments.

4.2 Parameter-Efficient Fine-Tuning (PEFT)

To address the challenges of model adaptation, Hu et al. (2022) introduced Low-Rank Adaptation (LoRA). Rather than retraining all model parameters, LoRA freezes the original weights and inserts trainable rank-decomposition matrices into the transformer layers. The authors proved that this method could match the performance of full fine-tuning while requiring only a

fraction of the trainable parameters. This is a cornerstone of the RetailBot Pro architecture; we employ LoRA adapters (rank 16) across the query and value projection layers (q proj, v proj) to achieve deep domain adaptation within the TinyLlama-1.1B model.

4.3 Quantized Fine-Tuning

Building upon the LoRA framework, Dettmers et al. (2023) developed QLoRA, which allows for the fine-tuning of models using 4-bit Normal Float (NF4) quantization. This breakthrough demonstrated that even massive 65B-parameter models could be tuned on a single 48GB GPU without losing performance. For this research, QLoRA is the primary technical enabler. By compressing the TinyLlama-1.1B memory footprint to approximately 1.1 GB, we successfully conducted fine-tuning within the hardware constraints of a Google Colab T4 GPU, proving that sophisticated retail AI development is possible on consumer-level resources.

4.4 Hallucination in Generative Models

Ji et al. (2023) conducted a comprehensive survey of hallucination in natural language generation, categorizing it into intrinsic and extrinsic types. Their findings underscore the critical importance of domain grounding and controlled generation parameters in high-stakes applications. RetailBot Pro addresses this through structured prompt engineering, low-temperature sampling (0.3), repetition penalties, and fine-tuning on curated instruction-response

pairs reflecting real retail support interactions.

4.5 Existing Frameworks and Tools

Tool / Framework	Role in RetailBot Pro
HuggingFace Transformers	Model loading, tokenizer, BitsAndBytesConfig
PEFT (Hu et al., 2022)	LoRA adapter configuration and model wrapping
TRL / Training Arguments	Supervised fine-tuning pipeline management
Bitext CS Dataset	Primary domain-specific training corpus
Gradio	Interactive chatbot frontend with public share
Google Colab T4 GPU	Hardware platform for model training

Table 1: Frameworks and Tools in RetailBot Pro

4.6 Gaps in Existing Literature

Despite extensive research on general-purpose LLMs and parameter-efficient tuning, several significant gaps remain in the retail AI space. First, most existing chatbot research focuses on general dialogue systems rather than the specific nuances of retail customer support terminology and policy structures. Second, few studies have demonstrated a complete QLoRA-based fine-tuning pipeline on a structured customer support dataset deployable on commodity hardware. RetailBot Pro aims to bridge

these gaps through a full-stack implementation and subsequent empirical analysis.

4.7 Contribution of This Study

This research provides a complete, reproducible, and hardware-accessible framework for domain-specific retail AI. Our primary contributions include:

- **A Domain Dataset:** A structured instruction-following corpus derived from the Bitext customer support dataset, formatted for retail dialogue fine-tuning.
- **Validated Pipeline:** A proven QLoRA fine-tuning workflow for TinyLlama-1.1B optimized for T4 GPU hardware.
- **Deployment Interface:** A production-ready Gradio chat-bot supporting real-time retail query resolution with integrated hallucination mitigation.
- **Empirical Data:** A comprehensive evaluation of the system's factual accuracy, response speed, and coherence dialogue on retail test scenarios.

V. METHODOLOGY

The development of RetailBot Pro follows a structured four-stage pipeline: dataset construction, model fine-tuning, hallucination mitigation, and Gradio deployment.

5.1 Dataset Construction

The training corpus was sourced from the publicly available **bitext/Bitext-customer-support-llm-chatbot-training-dataset** on HuggingFace. This dataset contains structured instruction-response pairs covering a comprehensive range of retail customer

support scenarios, including order management, return policies, shipping enquiries, refund procedures, and account management.

All training data is formatted using a consistent instruction-response prompt template:

```
### Instruction:
{instruction}
```

```
### Response:
{response}
```

To ensure training efficiency and prevent memory overrun on the T4 GPU, the dataset was sampled to **2,000 high-quality instruction-response pairs**, sufficient to achieve meaningful domain adaptation for the core retail support use cases.

5.2 QLoRA Fine-Tuning Configuration

We utilized **TinyLlama/TinyLlama-1.1B-Chat-v1.0** as the base model, loaded via BitsAndBytesConfig using 4-bit NF4 quantization with fp16 compute data type and double quantization enabled. This successfully compressed the model's VRAM requirements, making training feasible on the Google Colab T4 GPU (15 GB VRAM).

LoRA adapters are configured with rank $r=16$, scaling factor $\alpha=32$ ($\alpha/rank = 2.0$), and dropout rate 0.1, targeting the query and value projection matrices (q proj, v proj). This targeted adapter approach minimizes the number of trainable parameters while focusing adaptation on the attention layers most critical for dialogue quality.

5.2.1 Training Execution

Training was performed using the `TrainingArguments` and `Trainer` API from the HuggingFace `Transformers` library. Key hyperparameters were configured as follows: 2 training epochs, per-device batch size of 4, gradient accumulation steps of 4 (effective batch size of 16), learning rate of $2e-4$, fp16 mixed-precision training, and paged AdamW 8-bit optimizer. The model was saved at the end of each epoch using `save_strategy='epoch'`, and the complete training process on 2,000 samples was completed in approximately 20 minutes on the T4 GPU.

5.3 Hallucination Mitigation Strategy

RetailBot Pro implements a multi-layered hallucination mitigation strategy during inference. First, the system prompt explicitly constrains the model to its retail support role: *“You are a professional retail customer support assistant. Provide clear, polite and factual answers only.”* Second, generation is controlled with low temperature (0.3), top-p sampling (0.9), repetition penalty (1.2), and a maximum new token limit of 150 tokens. These constraints collectively steer the model toward factual, concise, policy-aligned responses.

5.4 Deployment Interface

For the user-facing component, we developed a Gradio `ChatInterface` with a soft theme, a descriptive title, and

four pre-configured retail example queries covering the most common support scenarios: order tracking, returns, shipping address changes, and refund policy enquiries. The interface is deployed via a publicly accessible Gradio share link, facilitating real-time testing and stakeholder demonstration.

VI. EXPERIMENTS AND RESULTS

6.1 Evaluation Objectives

The assessment of RetailBot Pro focused on five key performance indicators:

1. **Factual Accuracy:** Alignment with established retail policies and order management standards.
2. **Response Relevance:** The appropriateness and specificity of generated responses to retail queries.
3. **Response Latency:** The end-to-end time from query submission to generation.
4. **Dialogue Coherence:** The system’s ability to maintain context across multi-turn exchanges.
5. **Hallucination Mitigation:** The frequency of factually unsupported claims within the retail domain.

6.2 Experimental Setup

Testing was conducted using the Gradio interface deployed on Google Colab. We reloaded the fine-tuned TinyLlama-1.1B adapter and evaluated the system on a curated test set of 50 queries categorized into:

- Order Management (e.g., order status, cancellation).
- Returns and Exchanges (e.g., return

eligibility, exchange process).

- Refund Policies (e.g., refund timelines, partial refunds).
- Shipping and Delivery (e.g., address changes, delivery delays).
- Account and Payment (e.g., payment methods, billing queries).

6.3 Performance Metrics

Metric	Description	Result
Response Accuracy	Factual correctness rated against retail policies	~88% (44/50 correct)
Response Relevance	Proportion of on-topic responses by evaluators	~90%
Response Latency	Avg wall-clock time from query to response	1.5–2.8 seconds
Hallucination Rate	Proportion of responses with fabricated claims	~8% (4/50)
Multiturn Coherence	Follow-up queries resolved correctly	~84% (21/25)
User Satisfaction	Subjective rating 1–5 by 8 evaluators	4.3 / 5.0 avg

Table 2: RetailBot Pro Evaluation Results

6.4 Sample Query-Response Evaluations

Query	Outcome
Where is my or-	PASS – Policy-aligned response with

der?	tracking steps
How do I return a product?	PASS – Complete return procedure with label instructions
Can I change my shipping address?	PASS – Accurate response on change window and escalation
What is your refund policy?	PASS – Refund timelines and eligibility correctly described
How long does shipping take?	PASS – Standard and express timeframes correctly stated
Can you process my refund now?	PASS – Correctly deferred to escalation without fabricating confirmation

Table 3: Sample Query Evaluation Results

6.5 Training Convergence

Training loss decreased consistently across the 2-epoch training window, from approximately 2.61 at the initial logging step to 1.43 at the final training step, representing a 45.2% reduction. No significant overfitting was observed, consistent with the regularization provided by LoRA dropout (0.1) and the paged AdamW optimizer. Model checkpoints were saved at each epoch, and the best-performing checkpoint was retained for evaluation and deployment.

6.6 Limitations Observed

Several system limitations were identified during evaluation. The

current implementation does not incorporate a persistent knowledge retrieval layer (such as RAG), which means the model relies entirely on its fine-tuned parametric knowledge for responses. This makes it susceptible to hallucination for highly specific queries — such as exact return window durations for specific product categories — not adequately represented in the training corpus. Additionally, queries using highly informal phrasing occasionally produced suboptimal responses when colloquial terms did not align with the formal vocabulary of the training data. Future iterations should integrate a FAISS-based RAG pipeline to ground responses in verified policy documents.

6.7 Summary of Results

The experimental results collectively affirm that RetailBot Pro delivers reliable, retail-grounded conversational responses at production-grade latency. The combination of QLoRA fine-tuning for domain vocabulary adaptation and structured prompt engineering for hallucination control produces a system that substantially outperforms generic LLM responses on retail support content, achieves sub-3-second response latency suitable for real-time customer support, and maintains conversational coherence across multi-turn retail dialogue sequences.

VII. DISCUSSION

The findings of this study provide several important insights into the design and development of domain-specific

conversational AI systems for the retail sector. First, the results clearly demonstrate that QLoRA fine-tuning on a structured, domain-specific instruction dataset produces qualitatively superior retail customer support responses when compared to prompting a generic base language model. Through the fine-tuning process, the model becomes capable of internalizing retail-specific terminology, policy structures, operational workflows, and conversational response patterns that are typically underrepresented in general-purpose pre-training corpora. This domain adaptation significantly improves the system's ability to deliver responses that are not only accurate but also aligned with the tone, expectations, and conventions of customer service communication.

Second, the study highlights that the selection of LoRA target modules plays a measurable and meaningful role in determining the quality of domain adaptation. Specifically, targeting the `q_proj` and `v_proj` matrices — which correspond to the query and value projection components of the attention mechanism — concentrates the adapter learning process on the parts of the model most responsible for contextual understanding and response generation. Preliminary ablation experiments that utilized only `q_proj` adapters produced noticeably less coherent and less reliable policy-related responses. This outcome confirms that including `v_proj` in the adapter configuration provides additional contextual grounding and leads to more

stable and consistent outputs. These findings emphasize the importance of thoughtful adapter configuration when applying parameter-efficient fine-tuning techniques to specialized domains.

Third, the hallucination mitigation strategy implemented in this research — combining structured system prompts, low-temperature sampling, and repetition penalties — demonstrably reduces the frequency of fabricated or unsupported claims in retail support scenarios. By encouraging the model to remain conservative and grounded in the provided context, the system shows a marked reduction in unsupported policy statements and speculative answers. However, the remaining 8% hallucination rate highlights the ongoing importance of knowledge grounding and external data integration. Future research should therefore incorporate a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector search over retailer-specific documentation, such as return policies, shipping guidelines, and warranty terms. Such integration would allow the model to retrieve authoritative information dynamically, further improving factual accuracy and trustworthiness.

From an industry perspective, RetailBot Pro demonstrates that high-quality domain-specific conversational AI can be developed and deployed using accessible hardware and open-source tools. The complete training and deployment

pipeline runs on Google Colab's free NVIDIA T4 GPU tier in under 30 minutes, establishing a highly reproducible and cost-effective workflow. This result provides a practical template that can be adopted by retailers, e-commerce platforms, startups, and customer support teams that may not have access to specialized machine learning infrastructure. The accessibility of this approach significantly lowers the barrier to entry for adopting AI-driven customer support solutions.

Ethical considerations for production deployment remain an essential aspect of real-world implementation. Potential risks include the inadvertent embedding of policy inaccuracies present in the training dataset, the need for frequent model updates to reflect evolving return and refund policies, and the importance of maintaining transparent escalation pathways to human customer service agents for complex or sensitive interactions. Ensuring accountability, transparency, and regular system auditing will be critical for maintaining user trust. Future research should also explore persistent session memory architectures to enable long-term personalization, multilingual capabilities to support global customer bases, and integration with e-commerce platforms via secure API connectors to enable real-time access to order and shipping data. Collectively, these enhancements will further strengthen the practicality, scalability, and reliability of domain-specific conversational AI systems in retail and

beyond.

VIII. CONCLUSION

This research presents RetailBot Pro, a domain-specific conversational AI system designed specifically for the retail and e-commerce industry. The system was developed through QLoRA parameter-efficient fine-tuning of TinyLlama-1.1B-Chat using the Bitext customer support dataset, enabling the model to specialize in handling retail-focused conversations and customer service interactions. The training pipeline leveraged 4-bit NF4 quantization, allowing the model to be trained efficiently on a commodity NVIDIA T4 GPU while significantly reducing memory requirements and computational cost. Following training, the system was deployed as an interactive Gradio-based chatbot interface, providing a practical demonstration of how lightweight, fine-tuned large language models can be integrated into real-world applications. To enhance reliability and trustworthiness, the deployment incorporated hallucination mitigation strategies, ensuring that generated responses remain grounded, relevant, and aligned with customer support use cases.

Experimental evaluation was conducted using 50 structured test queries designed to simulate real retail customer interactions, including product inquiries, order tracking, return policies, and post-purchase support

scenarios. The results demonstrated strong performance across multiple evaluation metrics. RetailBot Pro achieved 88% response accuracy and 90% response relevance, indicating that the system consistently provided correct and contextually appropriate answers. In addition, the chatbot maintained sub-3-second response latency, confirming its suitability for real-time conversational deployment. The system recorded an 8% hallucination rate, reflecting the effectiveness of the implemented mitigation techniques, and achieved 84% multi-turn coherence, demonstrating its ability to sustain context across extended customer conversations. Collectively, these findings establish RetailBot Pro as a performant, efficient, and practically deployable conversational AI solution for retail environments.

The results of this work further confirm that parameter-efficient fine-tuning on structured, domain-specific datasets represents a highly effective and accessible paradigm for developing specialized AI systems. This approach significantly lowers the computational barrier to entry, enabling organizations and researchers to build tailored conversational agents without requiring large-scale infrastructure. The proposed system shows immediate applicability to retail customer support, e-commerce operations, and post-purchase service management, where fast, accurate, and context-aware responses are critical for

improving customer satisfaction and operational efficiency.

Future research directions will focus on expanding the capabilities of RetailBot Pro through the integration of Retrieval-Augmented Generation (RAG) to ground responses in company policy documents and knowledge bases. Additional enhancements include the introduction of persistent cross-session memory to enable long-term personalization, multilingual support to serve global customers, and voice interface integration for omnichannel accessibility. Further domain specialization can be achieved by fine-tuning the model on real customer support logs, allowing the system to continuously adapt to evolving customer needs and business requirements. The development blueprint established in this research is not limited to retail and can be directly transferred to other customer-facing domains such as logistics, hospitality, banking, and telecommunications, highlighting the broader impact and scalability of this approach.

APPENDIX

A. System Architecture Summary

RetailBot Pro integrates the following open-source components: fine-tuning via HuggingFace Transformers, PEFT, and TRL; generative inference via the fine-tuned TinyLlama adapter; and conversational UI via Gradio ChatInterface.

Component	Specification
Base Model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
Quantization	4-bit NF4, fp16 compute, double quantization
LoRA Configuration	r=16, alpha=32, dropout=0.1, q proj + v proj
Training Epochs	2 (effective batch 16, LR 2e-4, paged AdamW 8-bit)
Training Dataset	Bitext customer-support-llm-chatbot (2,000 samples)
Generative LLM	Fine-tuned TinyLlama-1.1B adapter
Generation Config	temp=0.3, top p=0.9, rep penalty=1.2, max new tokens=150
Chatbot Interface	Gradio ChatInterface with retail example queries
Training Platform	Google Colab T4 GPU (15 GB VRAM)
GitHub Repository	https://github.com/mayank021296/LLM-Chatbot-by-Using-Pretrained-Model

TABLE IV: RetailBot Pro System Specification

B. Training Configuration Code

```
training_args = TrainingArguments (
    output_dir="./retail-llm"
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4,
    num_train_epochs = 2, learning_rate = 2e-4,
    fp16 = True, logging_steps = 50,
```

```
save_strategy = "epoch", optim =  
"paged_adamw_8bit",  
dataloader_pin_memory = False  
)
```

<https://arxiv.org/abs/2401.02385>

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GitHub:

https://github.com/mayank021296/LLM_Chatbot_by_Using_Pretrained_Model

Acknowledgement

The authors acknowledge the open-source contributions of the HuggingFace, PEFT, TRL, and Gradio communities whose frameworks and tools form the technical foundation of RetailBot Pro. The Bitext customer-support-llm-chatbot-training-dataset, sourced from the HuggingFace Hub, provided the primary training corpus for this research.